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NEPHROLOGY – ORIGINAL PAPER

Development and validation of a multidimensional machine learning model for predicting delayed graft function in deceased donor kidney transplantation

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Abstract

Purpose Delayed graft function (DGF) remains a significant complication following deceased donor kidney transplantation. This study aimed to develop and validate a multidimensional machine learning model for predicting DGF by integrating clinical data, machine perfusion parameters, donor scores, and histopathological scores.

Methods A retrospective analysis was conducted on 961 deceased donor kidney transplant recipients from January 2019 to December 2021. The dataset was stratified by the target variable and randomly divided into training (80%) and independent testing (20%) cohorts. Fifteen data combinations across four dimensions were evaluated using six machine learning algorithms. Model performance was assessed using AUC, calibration curves, and decision curve analysis. SHAP analysis was employed for feature interpretation.

Results The optimal model combining clinical supplementary data with donor scores and histopathological scores using LightGBM achieved an AUC of 0.895 (95% CI 0.806–0.985) in the testing cohort, with accuracy of 89.1%, sensitivity of 59.1%, and specificity of 93.0%. Clinical data served as the foundational predictive dimension (AUC = 0.874), while histopathological scores provided significant incremental value (ΔAUC = 0.016). Among the top five ranked combinations, four included histopathological scores. Gain-based feature importance identified donor score, renal tubular necrosis, and total pathological score as the leading predictors, with pathological indicators occupying four of the top five positions; SHAP analysis further revealed blood urea nitrogen and donor score as the most influential individual features, with renal tubular necrosis being the top-ranked histopathological predictor.

Conclusion This multidimensional machine learning model demonstrates excellent predictive performance for DGF. The "scenario-model-threshold" clinical decision framework provides a practical tool for guiding donor kidney assessment and immunosuppressive therapy decisions in diverse clinical contexts.

Keywords Kidney transplantation · Delayed graft function · Machine learning · Prediction model · Multidimensional data fusion

Introduction

Kidney transplantation represents the optimal treatment for end-stage renal disease [1, 2]. Delayed graft function (DGF), defined as the requirement for dialysis within the first week posttransplantation, occurs in approximately 20–50% of deceased donor kidney transplants [3, 4]. DGF is associated with prolonged hospitalization, increased healthcare costs, and may adversely impact long-term graft survival [5, 6].

Existing DGF prediction models primarily rely on donor and recipient demographic characteristics without incorporating dynamic physiological parameters or histopathological findings [7, 8]. Machine learning algorithms have emerged as promising tools for clinical prediction [9, 10]. However, these studies have typically employed limited feature sets [11, 12].

To address these limitations, we developed and validated a comprehensive machine learning model integrating four

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data dimensions: clinical characteristics, machine perfusion parameters, donor quality scores, and histopathological evaluation scores. This study aimed to identify the optimal data combination and machine learning algorithm for DGF prediction and to establish the incremental value of each data dimension.

Materials and methods

Study design and population

This retrospective cohort study included 961 deceased donor kidney transplant recipients at The First Affiliated Hospital of Xi'an Jiaotong University between January 1, 2019 and December 31, 2021. DGF was defined as the requirement for dialysis within the first week posttransplantation. The cohort comprised 164 DBD (17.1%) and 797 DCD/DBCD (82.9%) donors, reflecting the composition of deceased organ donation in China. Recipients were eligible when complete four-dimensional data (clinical variables, machine perfusion parameters, donor score, and histopathological scores) were available; pediatric donors were excluded, as were recipients with primary nonfunction, graft loss or death within the first posttransplant week, or reoperation within the first week. All 961 kidneys (100%) underwent preimplantation biopsy and LifePort hypothermic machine perfusion. All recipients (100%) received rabbit antithymocyte globulin (ATG) induction therapy per center protocol during the study period; no recipient received IL-2 receptor monoclonal antibody induction. Maintenance immunosuppression followed a standard triple regimen.

Data collection and variable definitions

Data were extracted from four dimensions (Table 1):

Clinical data (28 variables): Donor and recipient demographics, comorbidities, ischemia times, and laboratory parameters including serum creatinine, blood urea nitrogen, and urine output.

Machine perfusion data (4 variables): Kidneys were preserved using the LifePort kidney transporter (Organ Recovery Systems, USA) with an initial pump pressure of 30 mmHg, and perfusion parameters (pressure, temperature, flow, resistance, and duration) were continuously recorded. Terminal flow and terminal resistance values were used for scoring, consistent with our previously validated HMP scoring model [12]. The four perfusion variables were derived from that model: flow score (0–5 points), resistance index score (0–8 points), time score (0–1 points, assigned 0 for HMP duration <12 h and 1 for ≥12 h), and total score (0–14 points, the sum of the three component scores).

Table 1 Description of 15 data combinations

Combination No	Data sources	Variables(*n*)	Description
1	Clinical	28	Recipient clinical indicators + donor basic information
2	Machine perfusion	4	LifePort perfusion parameters
3	Donor score	1	Chinese Donor Kidney Quality Score system
4	Histopathological score	15	Pathological scoring system based on Remuzzi criteria
5	Clinical + machine perfusion	32	Combination 1 + Combination 2
6	Clinical supplementary + donor score	21	Combination 1 (excluding donor score-related features) + Combination 3
7	Clinical + histopathological score	43	Combination 1 + Combination 4
8	Machine perfusion + donor score	5	Combination 2 + Combination 3
9	Machine perfusion + histopathological score	19	Combination 2 + Combination 4
10	Donor score + histopathological score	16	Combination 3 + Combination 4
11	Clinical supplementary + machine perfusion + donor score	25	Combination 1 (excluding donor score-related features) + Combination 2 + Combination 3
12	Clinical supplementary + donor score + histopathological score	36	Combination 1 (excluding donor score-related features) + Combination 3 + Combination 4
13	Clinical + machine perfusion + histopathological score	47	Combination 1 + Combination 2 + Combination 4
14	Machine perfusion + donor score + histopathological score	20	Combination 2 + Combination 3 + Combination 4
15	Clinical supplementary + machine perfusion + donor score + histopathological score	40	Combination 1 + Combination 2 + Combination 3 + Combination 4

Clinical supplementary data refers to the 20 remaining clinical variables after removing, from the original 28 clinical variables, the 8 variables corresponding to the 6 components of the donor scoring system

Donor score (1 variable): The donor score was the Chinese Donor Kidney Quality Score previously developed and validated in a multicenter cohort [11]. This composite score (range 0–49) integrates six donor-related components: donor age, cause of death, history of hypertension, terminal serum creatinine, duration of hypotension, and duration of cardiopulmonary resuscitation (CPR), with higher scores indicating greater DGF risk. To prevent information duplication between the donor score and its constituent variables, the clinical supplementary dataset was constructed by removing, from the 28 clinical variables, the 8 raw variables corresponding to these 6 components: donor age, primary disease (cause of death), hypertension history, terminal serum creatinine, the three systolic blood pressure tier variables (80–100, 50–80, and 0–50 mmHg, jointly representing hypotension duration), and CPR duration. The resulting clinical supplementary dataset comprised 20 variables (Supplementary Table S1). This design ensured that the donor score and the clinical supplementary variables carried non-overlapping information, allowing the model to evaluate their additive value.

Histopathological score (15 variables): Including glomerulosclerosis, interstitial fibrosis, tubular atrophy, and Remuzzi score based on preimplantation biopsy.

Immediately after graft procurement and completion of bench preparation, a core biopsy was obtained from the lower pole of each kidney using a 16-gauge Bard biopsy device (depth approximately 2 cm, including renal cortex and partial medulla). The puncture site was closed with an absorbable figure-of-eight suture to prevent perfusate leakage and ensure the accuracy of subsequent LifePort perfusion parameters. Specimens were paraffin-embedded and stained with HE, periodic acid-Schiff, and Masson trichrome, and were considered adequate when containing ≥7 glomeruli and ≥2 arterioles. All slides were evaluated by a single dedicated chief pathologist with 20 years of subspecialty experience in renal transplant pathology, using the Banff 2013 criteria [13], the Remuzzi score, and the Maryland Aggregate Pathology Index (MAPI), following a standardized protocol previously published and validated by our group [14]. Histopathological processing proceeded in parallel with hypothermic machine perfusion. In this retrospective study, organ allocation and discard decisions were based on comprehensive donor assessment, and histopathological results did not influence the timing of transplantation; consequently, the biopsy procedure did not prolong cold ischemia time in any case, including nighttime procedures.

Fifteen data combinations were constructed: four single-dimension combinations, six two-dimension combinations, four three-dimension combinations, and one four-dimension combination (Table 1).

Data preprocessing

Missing values (0.8%–7.3% across features) were imputed using multiple imputation by chained equations (MICE) [15, 16] with ten iterations and five imputations (predictive mean matching for continuous variables and Logistic Regression for categorical variables). Continuous variables were standardized using Z score normalization. Categorical variables were encoded using one-hot encoding.

Feature selection employed a multi-step strategy: univariate screening (t test/Chi-square, $P<0.1$), correlation analysis (excluding $|r|>0.8$), recursive feature elimination (RFE) [17, 18], and SHAP value analysis [19].

Machine learning models

Six machine learning algorithms were evaluated: Logistic Regression (LR) [20], Support Vector Machine (SVM) [21], Random Forest (RF) [22], Extra Trees (ET) [23], XGBoost [24], and LightGBM [24]. Hyperparameters were optimized using grid search with fivefold cross-validation (Table 2).

To address class imbalance (DGF incidence 11.2%), class weighting strategies were applied: class_weight='balanced' for tree-based models, and scale_pos_weight=7.9 for XGBoost and LightGBM.

Model evaluation

The dataset was stratified by the target variable and randomly partitioned into a training set (80%) and an independent testing set (20%, $n = 193$), with the random state

Table 2 Hyperparameter search space for machine models

Model	Hyperparameter	Search range
LR	C	0.001, 0.01, 0.1, 1, 10, 100
LR	Penalty	l2
SVM	C	0.1, 1, 10, 100
SVM	Gamma	scale, auto, 0.001, 0.01, 0.1
RF	n_estimators	100, 200, 300, 500
RF	max_depth	3, 5, 7, 10, None
RF	min_samples_split	2, 5, 10
ET	n_estimators	100, 200, 300, 500
ET	max_depth	3, 5, 7, 10, None
XGBoost	n_estimators	100, 200, 300
XGBoost	max_depth	3, 5, 7
XGBoost	learning_rate	0.01, 0.05, 0.1
LightGBM	n_estimators	100, 200, 300
LightGBM	max_depth	3, 5, 7, − 1
LightGBM	learning_rate	0.01, 0.05, 0.1

LR Logistic Regression; *SVM* Support Vector Machine; *RF* Random Forest; *ET* Extra Trees

fixed at 42. Hyperparameter optimization and model selection were performed using fivefold stratified cross-validation on the training set; the testing set was reserved exclusively for final performance evaluation. Model performance was evaluated using area under the receiver operating characteristic curve (AUC), accuracy, precision, sensitivity, specificity, F1 score, and Youden index.

Decision curve analysis (DCA) assessed clinical utility by calculating net benefit across different threshold probabilities. Calibration was evaluated using calibration curves and Hosmer–Lemeshow test.

Statistical analysis

Statistical analysis was performed using Python 3.9 with scikit-learn 1.0.2 and SHAP 0.40.0. Continuous variables were expressed as mean ± standard deviation. Categorical variables were expressed as percentages. Between-group comparisons used independent t tests for continuous variables and Chi-square tests for categorical variables. DeLong test compared AUC differences between models. Two-sided $P<0.05$ was considered statistically significant.

Results

Study population characteristics

The study included 961 deceased donor kidney transplant recipients, of whom 587 (61.1%) were male and 374 (38.9%) were female. DGF occurred in 108 patients (11.2%). Table 3 presents the baseline characteristics stratified by DGF status. DGF-positive patients had significantly higher donor age (53.6±16.2 vs 41.2±14.8 years, $P<0.001$), higher prevalence of donor hypertension (53.7% vs 36.8%, $P<0.001$), longer cold ischemia time (11.8±4.2 vs 10.2±3.6 h, $P<0.001$), higher total pathological score (6.2±3.1 vs 3.8±2.4, $P<0.001$), and higher resistance index (0.34±0.14 vs 0.27±0.10, $P<0.001$).

Model performance comparison

Table 4 presents the performance of 15 data combinations across 6 machine learning algorithms. The top-performing configuration was "clinical supplementary + donor score + pathological score" using LightGBM, achieving an AUC of 0.895 (95% CI 0.806–0.985) in the testing cohort, with accuracy of 89.1%, sensitivity of 59.1%, specificity of 93.0%, precision of 52.0%, and F1 score of 55.3%.

Table 3 Baseline characteristics of study population

Characteristic	DGF-negative (n=853)	DGF-positive (n=108)	P value
Recipient age (years)	45.1±12.4	46.8±13.2	0.182
Recipient male (%)	60.5%	66.7%	0.198
Dialysis duration (months)	27.8±17.6	33.2±21.4	0.008
Preoperative SCr (micromol/L)	785.4±234.6	912.3±287.5	<0.001
Preoperative BUN (mmol/L)	18.6±6.2	22.4±7.8	<0.001
Donor age (years)	41.2±14.8	53.6±16.2	<0.001
Donor male (%)	58.0%	59.3%	0.792
Donor type, n (%)	DBD 164 (17.1%); DCD/DBCD 797 (82.9%)		–
Donor hypertension (%)	36.8%	53.7%	<0.001
Donor terminal SCr (micromol/L)	85.2±38.6	112.4±52.3	<0.001
CPR duration (min)	12.5±8.4	18.6±12.3	<0.001
Hypotension episode (%)	28.4%	45.4%	<0.001
Cold ischemia time (h)	10.2±3.6	11.8±4.2	<0.001
Warm ischemia time (min)	8.5±4.2	10.2±5.6	0.002
Perfusion flow (mL/min)	93.2±30.8	82.4±28.6	0.001
Resistance index	0.27±0.10	0.34±0.14	<0.001
Total pathological score	3.8±2.4	6.2±3.1	<0.001
Acute pathological score	1.9±1.2	2.4±1.4	0.001
Chronic pathological score	1.8±1.4	3.6±2.2	<0.001
Banff score	2.8±1.6	4.2±2.1	<0.001
RI score	1.8±1.2	2.8±1.5	<0.001

Data are expressed as mean ± standard deviation or percentage. SCr serum creatinine; *BUN* blood urea nitrogen; *CPR* cardiopulmonary resuscitation; *RI* resistance index

Table 4 Performance of top ten data combinations in testing cohort

Rank	Data combination	Algorithm	AUC	95% CI	Accuracy	Sensitivity	Specificity	Precision
1	Clinical supplementary + donor score + pathological	LightGBM	0.895	0.806–0.985	0.891	0.591	0.930	0.520
2	clinical + machine perfusion + pathological	Random Forest	0.890	0.799–0.981	0.902	0.636	0.936	0.560
3	Clinical + pathological	Random Forest	0.890	0.799–0.981	0.902	0.636	0.936	0.560
4	Donor score + pathological	LightGBM	0.876	0.781–0.972	0.881	0.545	0.924	0.480
5	Clinical	XGBoost	0.874	0.778–0.971	0.917	0.682	0.947	0.625
6	Clinical supplementary + donor score	Extra Trees	0.863	0.764–0.963	0.886	0.500	0.936	0.500
7	Machine perfusion + donor score + pathological	Random Forest	0.840	0.735–0.946	0.922	0.545	0.971	0.706
8	Clinical supplementary + machine perfusion + donor score + pathological	Random Forest	0.840	0.735–0.946	0.922	0.545	0.971	0.706
9	Clinical + machine perfusion	Random Forest	0.834	0.727–0.942	0.917	0.591	0.959	0.650
10	Clinical supplementary + machine perfusion + donor score	Extra Trees	0.816	0.705–0.927	0.886	0.455	0.942	0.500

The second- and third-ranked configurations were "clinical + machine perfusion + pathological score" and "clinical + pathological score" using Random Forest, both achieving AUC of 0.890 (95% CI 0.799–0.981).

Notably, clinical data alone using XGBoost achieved an AUC of 0.874 (95% CI 0.778–0.971), with the highest sensitivity (68.2%) and specificity (94.7%) among single-dimension models.

Incremental value of data dimensions

The incremental predictive value of each data dimension was evaluated by comparing AUC changes:

Clinical data: Served as the foundational predictive dimension with AUC = 0.874 (XGBoost).

Histopathological scores: Provided significant incremental value. Adding pathological scores to clinical data increased AUC from 0.874 to 0.890 (Delta AUC = +0.016).

Machine perfusion parameters: Showed limited independent contribution. Adding machine perfusion to clinical data increased AUC from 0.874 to 0.834 (Delta AUC = −0.040), likely due to collinearity with clinical variables.

Donor scores: Moderate incremental value. Combining donor scores with clinical supplementary data achieved AUC = 0.863 (Delta AUC = +0.011 from clinical supplementary alone).

Top-performing model configurations

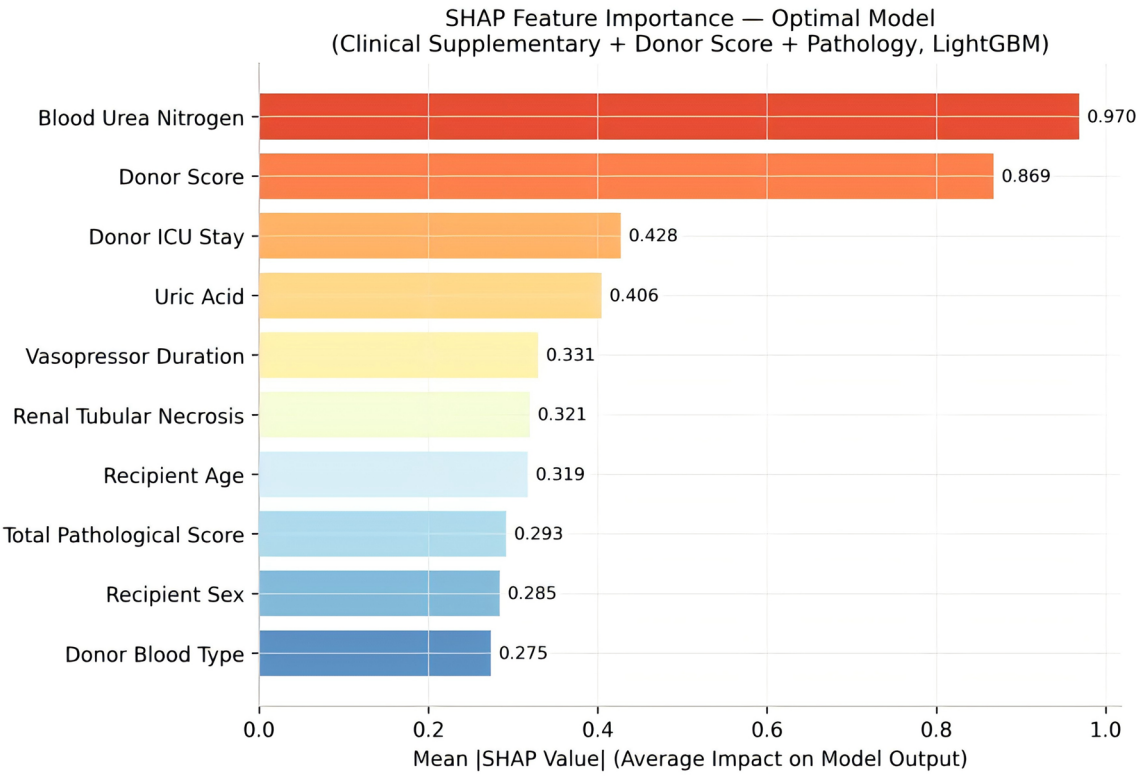
Table 5 presents the top ten model configurations ranked by testing AUC. The top five configurations all incorporated histopathological scores, confirming their significant predictive contribution. LightGBM and Random Forest consistently outperformed other algorithms across multiple data combinations.

Feature importance analysis

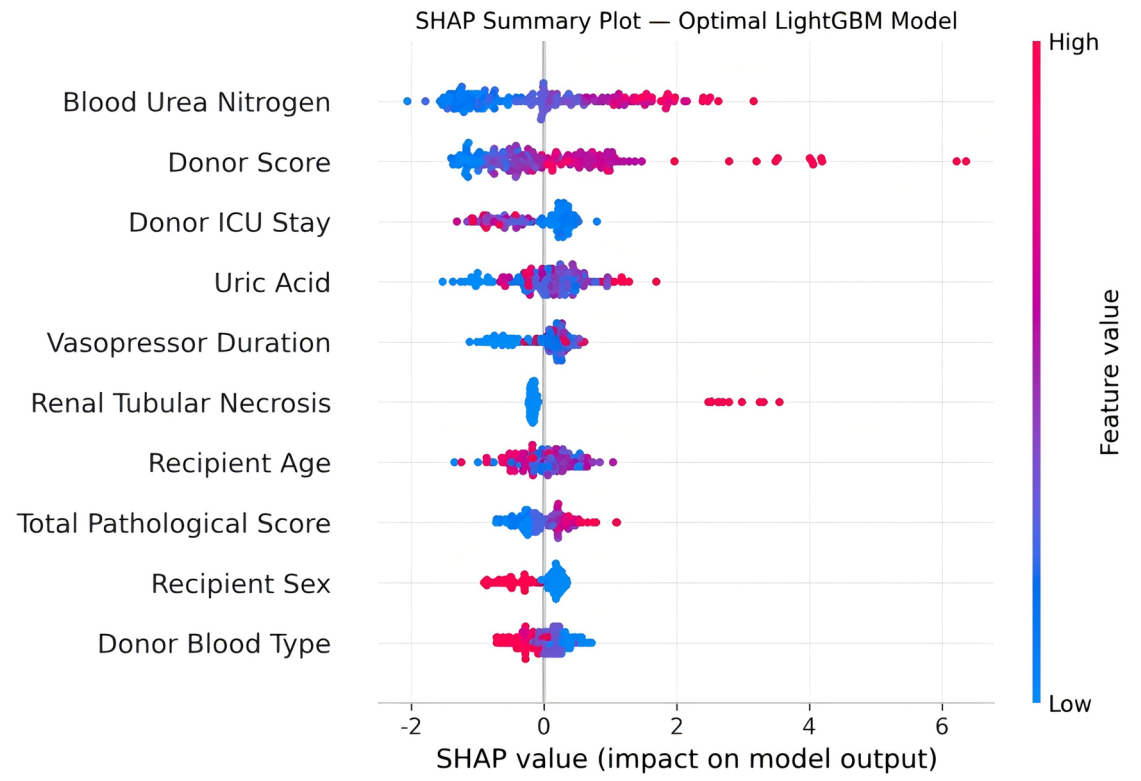
Feature importance of the optimal LightGBM model was assessed using two complementary approaches. Gain-based

Table 5 Top ten model configurations ranked by testing AUC

Rank	Data combination	Algorithm	AUC	95% CI	Sensitivity	Specificity
1	Clinical supplementary + donor score + pathological	LightGBM	0.895	0.806–0.985	0.591	0.930
2	Clinical + machine perfusion + pathological	Random Forest	0.890	0.799–0.981	0.636	0.936
3	Clinical + pathological	Random Forest	0.890	0.799–0.981	0.636	0.936
4	Donor score + pathological	LightGBM	0.876	0.781–0.972	0.545	0.924
5	Clinical	XGBoost	0.874	0.778–0.971	0.682	0.947
6	Clinical supplementary + donor score	Extra Trees	0.863	0.764–0.963	0.500	0.936
7	Machine perfusion + donor score + pathological	Random Forest	0.840	0.735–0.946	0.545	0.971
8	Clinical supplementary + machine perfusion + donor score + pathological	Random Forest	0.840	0.735–0.946	0.545	0.971
9	Clinical + machine perfusion	Random Forest	0.834	0.727–0.942	0.591	0.959
10	Machine perfusion + pathological	Random Forest	0.800	0.685–0.914	0.455	0.959


SHAP Feature Importance — Optimal Model
(Clinical Supplementary + Donor Score + Pathology, LightGBM)
Blood Urea Nitrogen
Donor Score
Donor ICU Stay
Uric Acid
Vasopressor Duration
Renal Tubular Necrosis
Recipient Age
Total Pathological Score
Recipient Sex
Donor Blood Type
0.970
0.869
0.428
0.406
0.331
0.321
0.319
0.293
0.285
0.275
0.0
0.2
0.4
0.6
0.8
1.0
Mean |SHAP Value| (Average Impact on Model Output)

(A)


SHAP Summary Plot — Optimal LightGBM Model
Blood Urea Nitrogen
Donor Score
Donor ICU Stay
Uric Acid
Vasopressor Duration
Renal Tubular Necrosis
Recipient Age
Total Pathological Score
Recipient Sex
Donor Blood Type
-2
0
2
4
6
SHAP value (impact on model output)
High
Feature value
Low

(B)

◂**Fig. 1** SHAP analysis of the optimal LightGBM model (clinical supplementary data + donor score + histopathological scores). **A** Feature importance ranked by mean absolute SHAP value. **B** SHAP value distribution for the top ten features; red indicates higher feature values and blue indicates lower feature values

feature importance identified donor score (relative importance 0.182), renal tubular necrosis (0.156), total pathological score (0.128), chronic pathological score (0.098), and glomerular mesangial proliferation (0.087) as the top five features, with pathological indicators occupying four of the five positions, underscoring the contribution of histopathological assessment. SHAP analysis further quantified each feature's average impact on model output (Fig. 1): blood urea nitrogen (mean |SHAP|= 0.970) and donor score (0.869) ranked as the most influential individual predictors, followed by donor ICU stay (0.428), uric acid (0.406), and vasopressor duration (0.331); renal tubular necrosis (0.321) was the highest-ranked histopathological feature. Notably, donor terminal serum creatinine was not included as a standalone feature in this combination, as it constitutes a component of the donor score; blood urea nitrogen therefore served as the representative independent biochemical marker of donor renal function.

Figure 1 presents the SHAP analysis results for the optimal LightGBM model (clinical supplementary data + donor score + histopathological scores), showing (A) feature importance ranked by mean absolute SHAP value and (B) the SHAP value distribution for the top ten features, demonstrating the direction and magnitude of each feature's impact on model predictions.

Clinical utility analysis

Decision curve analysis (Fig. 2A) demonstrated that the optimal multidimensional model (LightGBM) provided higher net benefit than the clinical-only model (XGBoost) across threshold probabilities of 0.05–0.15, making it suitable for DGF risk screening. The clinical-only model showed stable net benefit across threshold probabilities of 0.15–0.50, suitable for routine preoperative assessment.

Clinical impact curves (Fig. 2B) showed that at a threshold probability of 0.10, the multidimensional model identified approximately 15 true positive cases per 100 patients, compared to 12 for the clinical-only model.

The confusion matrices of the two principal models on the independent test set (n = 193) are presented in Fig. 3: the multidimensional LightGBM model yielded TP 13, FN 9, FP 12, and TN 159 (sensitivity 59.1%, specificity 93.0%, negative predictive value 94.6%), whereas the clinical-only XGBoost model yielded TP 15, FN 7, FP 9, and TN 162 (sensitivity 68.2%, specificity 94.7%, negative predictive value 95.9%).

Discussion

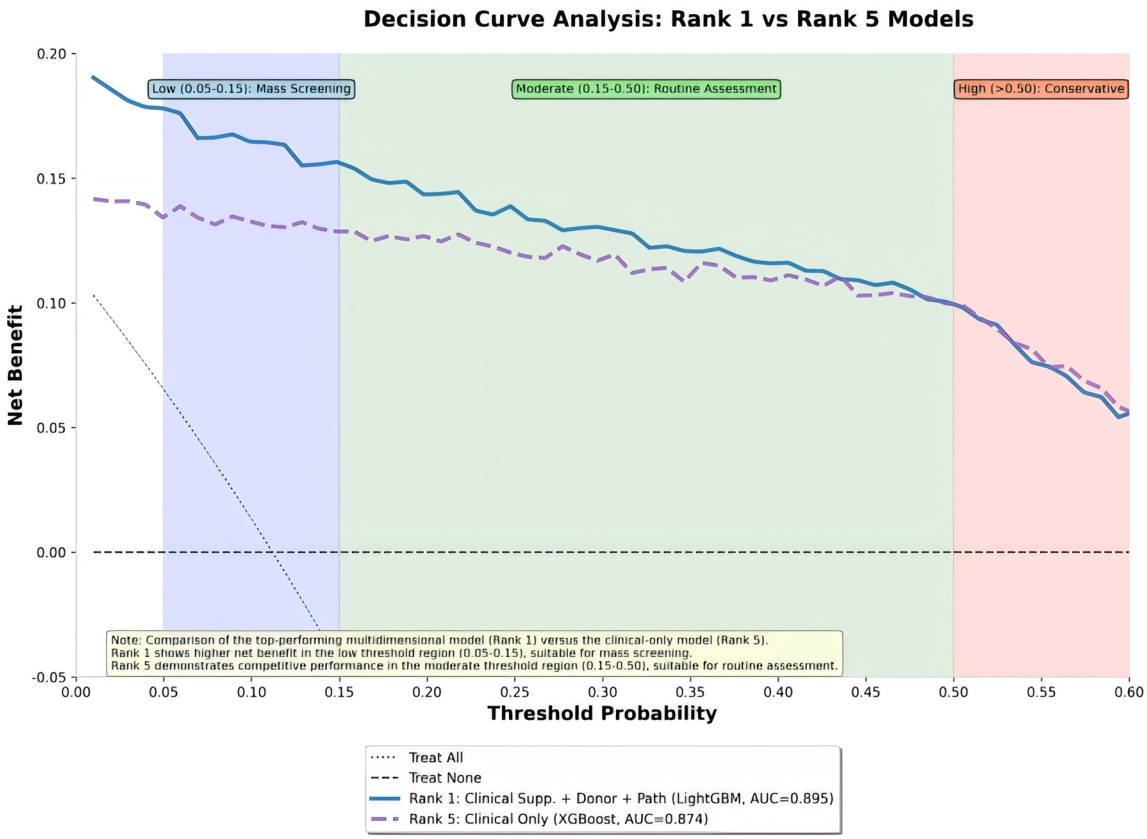
This study developed and validated a multidimensional machine learning model for predicting DGF in deceased donor kidney transplantation. The optimal model, integrating clinical supplementary data with donor scores and histopathological scores using LightGBM, achieved an AUC of 0.895, representing a 2.4% improvement over clinical data alone. Notably, the 20–50% DGF incidence reported in the Introduction derives predominantly from Western cohorts, whereas Chinese multicenter data report considerably lower incidences under the same definition [4]; DGF incidence at our center varied between 11.2% and 27.0% across study periods.

Principal findings

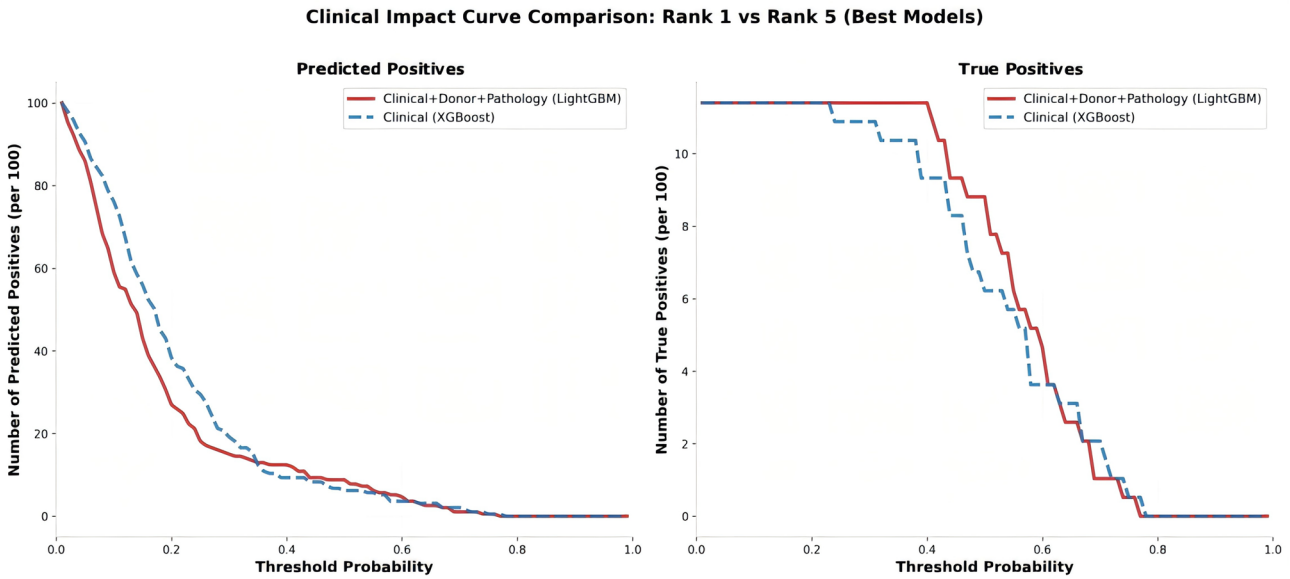
Clinical data as foundational contribution: Clinical data demonstrated the strongest independent predictive capability (AUC = 0.874), establishing its role as the "gatekeeper" in donor kidney quality assessment. Core predictive features including serum creatinine, blood urea nitrogen, donor age, and hypertension history enable rapid initial risk stratification, particularly valuable in time-constrained scenarios.

Histopathological scores provide significant incremental value: Pathological scores demonstrated substantial incremental predictive contribution (ΔAUC = 0.016): gain-based feature importance placed four pathological indicators among the top five features, and SHAP analysis identified renal tubular necrosis as the leading histopathological predictor—consistent with our previous finding that tubular injury/necrosis is an independent risk factor for DGF [14]. However, traditional pathological scoring has inherent limitations. Although the single-reader design of this study eliminated interobserver variability within our dataset, semi-quantitative scoring shows only moderate interobserver agreement in the literature (κ = 0.4–0.6) [25, 26], which may limit reproducibility across readers and centers. Moreover, histopathological assessment requires tissue processing that precludes real-time application (turnaround >24 h for paraffin-embedded sections, and 45–60 min even for frozen sections in prospective settings); in our retrospective cohort, biopsies were processed in parallel with machine perfusion and did not prolong cold ischemia time. Semi-quantitative simplification also entails information loss.

The top predictors identified by SHAP are biologically coherent. Renal tubular necrosis, the highest-ranked histopathological feature, is the direct histological signature of ischemia–reperfusion injury, the core mechanism of



(A)



(B)

Fig. 2 Clinical utility analysis of optimal models. **A** Decision curve analysis comparing the top-performing multidimensional model (clinical supplementary + donor score + pathological score, LightGBM) versus the clinical-only model (clinical data, XGBoost). The multidimensional model provides higher net benefit at threshold probabilities of 0.05–0.15, while the clinical-only model shows stable net benefit at threshold probabilities of 0.15–0.50. **B** Clinical impact curves showing predicted and true positive cases per 100 patients at different threshold probabilities. At threshold probability of 0.10, the multidimensional model identifies approximately 15 true positive cases per 100 patients, compared to 12 for the clinical-only model

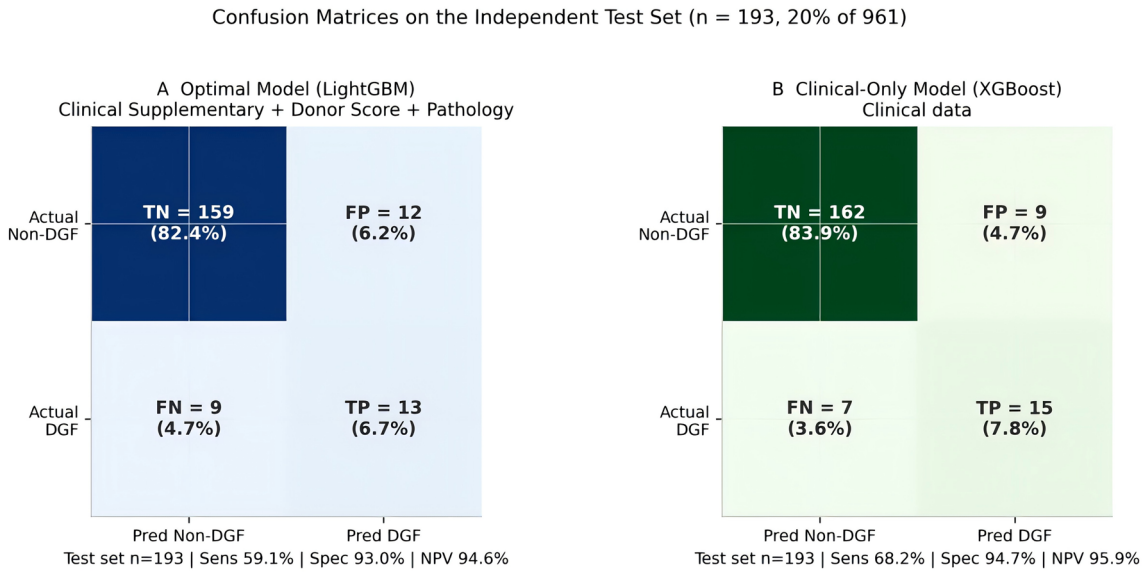


Fig. 3 Confusion matrices of the two principal models on the independent test set (n = 193). **A** Optimal multidimensional model (clinical supplementary + donor score + histopathological scores, LightGBM): TP 13, FN 9, FP 12, TN 159. **B** Clinical-only model (XGBoost): TP 15, FN 7, FP 9, TN 162. The test set size (n = 193) corresponds to 20% of the 961 cases under the verified 80%/20% stratified partition

DGF, integrating ischemic, hemodynamic, and preservation-related insults from donor maintenance through storage; this aligns with our previous finding that tubular injury/necrosis independently predicts DGF [14]. The donor score aggregates six established donor-risk domains and thus captures the baseline injury burden of the graft beyond any single variable. Blood urea nitrogen, the top individual predictor, reflects donor renal function together with prerenal perfusion, catabolic state, and volume status during ICU maintenance—complementary dimensions to creatinine, which was excluded as a donor score constituent.

Machine perfusion parameters show synergistic enhancement: Machine perfusion parameters exhibited "independent weakness but synergistic enhancement" characteristics. While single-dimension performance was modest (best AUC 0.611), integration with clinical and pathological data demonstrated good synergistic effects through capturing complex nonlinear interactions.

Comparison with existing studies

Our optimal model AUC of 0.895 compares favorably with existing DGF prediction models. Kawakita et al. [9] reported machine learning models with AUC of 0.735 using UNOS data. Jen et al. [10] achieved AUC of 0.759 with automated machine learning approaches. Zheng et al. [27] from our center previously reported AUC of 0.89 combining donor scores, pathological scores, and HMP resistance index in 333 cases. Our study validates and extends this finding with larger sample size (961 cases) and systematic comparison of 15 data combinations and 6 algorithms.

Our group previously developed an HMP-based scoring model for DGF prediction using perfusion variables alone ([12]; c-statistic 0.712), with the prognostic value of perfusion parameters further validated in large cohorts and meta-analyses [28–30]; that model served as the foundation of the machine perfusion dimension in the present study. The current work extends that single-dimension approach in scope (4 data dimensions across 961 transplants), methodology (6 machine learning algorithms across 15 data combinations), and clinical deliverable (a scenario-model-threshold framework rather than a single fixed score), improving predictive performance from 0.712 to 0.895.

The donor score used in this study is a Chinese population-specific instrument developed and validated in a 29-center cohort [11]; in our dataset, it achieved an AUC of 0.643 as a standalone predictor. This discrimination is comparable to that reported for internationally used scores: KDPI shows a c-statistic of approximately 0.60 for graft outcomes, and external validation of existing DGF scores yielded AUCs of 0.51–0.69 [31]. More recently, the Canadian Anatomic Kidney Score (CAKS), a macroscopic anatomic assessment, predicted 1-year eGFR <45 ml/min with an AUC of 0.63 [32]. These findings collectively indicate that single donor quality scores provide only modest discrimination, underscoring the rationale for multidimensional machine learning integration.

Clinical implications

The model's high specificity (93.0%) but relatively lower sensitivity (59.1%) suggests its positioning as a "high-specificity risk warning tool" rather than a high-sensitivity screening tool. When the model predicts high DGF risk, the positive predictive value of 52.0% should trigger intensified monitoring and individualized immunosuppressive management, such as delayed CNI initiation [33]. In centers where induction therapy is not universal, ATG induction may be considered for model-identified high-risk recipients; however, current evidence indicates that ATG does not reduce DGF incidence per se [34, 35], its principal value lying in reducing rejection risk during the DGF period and in high-immunological-risk recipients [36–38]. Conversely, the negative predictive value of 94.6% indicates that a negative prediction reliably identifies kidneys suitable for standard allocation. We explicitly report the sensitivity–specificity trade-off—the multidimensional LightGBM model achieved an AUC of 0.895 with a sensitivity of 59.1%, whereas the clinical-only XGBoost model achieved an AUC of 0.874 with a sensitivity of 68.2% (Fig. 3)—so that clinicians can prioritize case finding or overall discrimination according to the clinical scenario.

Based on decision curve analysis, we propose scenario-specific application strategies: (1) For large-scale screening, use the multidimensional model with low threshold (0.05–0.15); (2) for routine preoperative assessment, use clinical-only model with moderate threshold (0.15–0.50); (3) for marginal donor evaluation, use multidimensional model with individualized threshold (0.30–0.60).

Limitations

This study has several limitations. First, the single-center retrospective design limits generalizability; although our cohort reflects the full spectrum of Chinese deceased donor practice (82.9% DCD/DBCD), external validation—particularly in populations with higher DGF incidence and longer ischemia times—is required before the model can be generalized. A multicenter prospective external validation study is planned, incorporating long-term endpoints (1-year eGFR), prospective quantification of interobserver pathological agreement, and standardized real-time pathology workflows. Second, histopathological scoring was performed by a single dedicated transplant pathologist. Although this design eliminated interobserver variability within our dataset, it precluded internal quantification of interobserver agreement; published data indicate that agreement for preimplantation biopsy scoring is only moderate even among specialists (κ = 0.4–0.6) [25, 26], and reproducibility across readers and centers remains to be established in external validation. Third, the retrospective design may introduce selection bias and confounding factors, and DGF-positive cases were limited (n = 108), which may constrain model training for complex machine learning algorithms. Fourth, long-term outcomes such as 1-year eGFR and graft survival were not analyzed, as systematic long-term follow-up data were not collected in this retrospective cohort; these endpoints are pre-specified in our planned prospective multicenter validation study. Fifth, the uniform induction protocol (ATG in all recipients) precluded evaluation of induction strategies in relation to DGF.

Future directions

Future research should focus on: (1) multi-center external validation to assess generalizability; (2) prospective clinical trials to evaluate real-world effectiveness; (3) integration with automated pathological image analysis to reduce subjectivity and processing time; (4) development of user-friendly clinical decision support systems; (5) extension to predict long-term graft outcomes beyond DGF, with 1-year eGFR and graft survival as pre-specified endpoints of our planned prospective multicenter validation study.

Conclusion

This study developed and validated a multidimensional machine learning model for predicting DGF in deceased donor kidney transplantation. The optimal model integrating clinical supplementary data, donor scores, and histopathological scores using LightGBM achieved excellent predictive performance (AUC = 0.895). Clinical data provided the foundational predictive contribution, while histopathological scores offered significant incremental value. The model provides a robust tool for preoperative risk stratification, with scenario-specific application strategies based on decision curve analysis supporting clinical implementation.

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Data availability The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy restrictions.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Research involving human participants This retrospective study was approved by the Institutional Review Board of The First Affiliated Hospital of Xi'an Jiaotong University (Approval Number: XJTU-1AF2024LSYY-347) and conducted in accordance with the Declaration of Helsinki.

Informed consent Informed consent was obtained from all participants (or their legal guardians) prior to their inclusion in the study.

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